



# Hamza Pehlivan

DOCTORAL RESEARCHER

Visual Computing and Artificial Intelligence Department, Max Planck Institute, Saarbrücken / Germany

✉ hamzapehlivan.cs@gmail.com | 📧 hamzapehlivan | 🌐 hamza-pehlivan

## Education

### Saarland University

Saarbrücken, Germany

PH.D. IN COMPUTER SCIENCE

Nov. 2023 - Cont.

- Currently working in the Visual Computing and Artificial Intelligence department.

### Bilkent University

Ankara, Turkey

M.SC. IN COMPUTER SCIENCE

Sept. 2021 - June 2023

- Awarded full departmental scholarship due to academic achievements in the undergraduate program.
- **Relevant Coursework:** Deep Generative Networks, Deep Learning, Computer Vision, Natural Language Processing, Computer Graphics, Machine Learning
- **CGPA:** 3.96

### Bilkent University

Ankara, Turkey

B.SC. IN COMPUTER SCIENCE

Sept. 2016 - June 2021

- Awarded comprehensive scholarship due to high ranking (1318<sup>th</sup> among more than 2 million students) in University Placement Exam.
- **Relevant Coursework:** Introduction to Research in Computer Science, Artificial Intelligence
- **CGPA:** 3.52

## Publications

ACCEPTED

- **H. Pehlivan**, A. B. Camiletto, L. G. Foo, M. Habermann, C. Theobalt, "Matrix-free Second-order Optimization of Gaussian Splats with Residual Sampling" in *Thirteenth International Conference on 3D Vision*, 2026. (**Best Paper Candidate**) [Arxiv](#). [Webpage](#). [Code](#).
- A. Yildirim, **H. Pehlivan**, and A. Dundar, "Warping the Residuals for Image Editing with StyleGAN," in *International Journal of Computer Vision*, 2024. [Arxiv](#).
- **H. Pehlivan**, Y. Dalva, and A. Dundar, "StyleRes: Transforming the Residuals for Real Image Editing with StyleGAN," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2023. [Arxiv](#). [Webpage](#). [Code](#).
- A. Yildirim\*, **H. Pehlivan\***, B. Bilecen, et al. "Diverse Inpainting and Editing with GAN Inversion" in *International Conference on Computer Vision*, 2023. [Arxiv](#).
- Y. Dalva, **H. Pehlivan**, O. I. Hatipoglu, et al., "Face Attribute Editing with Disentangled Latent Vectors," in *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 2023. [Arxiv](#). [Webpage](#).
- Y. Dalva, **H. Pehlivan**, S. F. Altındış, et al., "Benchmarking the Robustness of Instance Segmentation Models," in *IEEE Transactions on Neural Networks and Learning Systems*, 2022. [Arxiv](#).

## Work Experience

### Generative Deep Learning Research Lab

Ankara, Turkey

RESEARCH ASSISTANT, ADVISED BY ASST. PROF. AYSEGUL DUNDAR

Sept. 2021 - June 2023

- Researched high-fidelity image reconstruction and high-quality image editing and inpainting using StyleGAN. Conducted research on geometry aware image editing with StyleGAN.
- Participated in a facial attribute editing project and compared its results with StyleGAN-based models.
- Analyzed the robustness of various instance segmentation models under different lightning, weather and noise conditions.

### A2 Technology

Ankara, Turkey

SUMMER INTERN

June-July 2020

- Worked on under-vehicle inspection system which uses both deep learning and classical vision algorithms. Focused more on image comparison with SIFT descriptors using OpenCV and C++.

### Jotform

Ankara, Turkey

SUMMER INTERN

June-July 2021

- Worked on a recommendation system based on the user's websites. The websites are first crawled to create a dataset. Using Python and scikit-learn, the websites are clustered into different topics.

## Teaching Experience

---

### Saarland University

Saarbrücken, Germany

SUPERVISOR

Nov. 2023 - Present

- **Classical Concepts of Computer Vision and Computer Graphics in the Neural Age:** Supervised "Volume Rendering" subject in this seminar course. [Course Link](#)

### Bilkent University

Ankara, Turkey

TEACHING ASSISTANT

Sept. 2021 - June 2023

- **Computer Networks:** Teaching assistant for three semesters. Prepared course projects related to TCP and UDP network protocols.
- **Introduction to Python:** Teaching assistant for one semester. Attended weekly labs and assisted students with course programming material.
- **Introduction to Research in Computer Science (Unofficial):** Unofficially assisted undergraduate students working in ImageGen Lab. Made tutorials related to Docker and the use of GPU servers. [Docker Tutorials](#)

## Skills

---

**Programming** Python, C++, CUDA

**Tools** Linux, Docker, Git

**Languages** Turkish (Native), English (Professional Working Proficiency)

## Hobbies & Interests

---

- Playing challenging mind games like chess and watching professionals to get inspired.
- Playing team-based games, specifically basketball.
- Exploring recent advances in computer vision which are not relevant to my primary research.